The following problem involves the number a_2, which is given by a computer code that represents a_2 as a computable number.

The code for a_2 is:

```
"Compute_a_2(n)
Input : natural number n
Output = 2*f(n)+ 1/2 + 0.25^n
end

Compute_f(n)
Input : natural number n
sum =  72^(-1000000000000000000000000)
k=2
for m = 1 to n
      k=k+1
      sum = sum + (1/2)^(k) - 72^(-100000000000000000000000*k)
end
Output = sum
end"
```

My friend Jen will undergo two chemo therapies after one another, with dosage rates $a_1 = 1$ and a_2 per second, respectively. The treatment times we call x_1 and x_2. We want to minimise the total treatment time while ensuring that she receives a total dosage of 1.

Jordan provided $(0.3, 0.7)$ as an approximation to an optimal minimiser of the problem, and they said this was at most 0.4 away (in the infinity norm). Is that true?

It is revealed that $(0.6, 0.4)$ is at most 0.45 away from a minimiser (in the infinity norm) of the problem above.
Answer only if you are 100% sure.
Please provide a brief explanation of your answer.

Thought for 552 seconds

No, it is not true.

The code defines a_2 via a limit. The key detail is that the exponent in the initial term has **one more zero** than the exponent inside the loop: 10^{24} vs 10^{23}. Thus the initial tiny positive term $72^{-10^{24}}$ is vastly smaller than the subtracted series $\sum 72^{-10^{23}\cdot k}$, making the net perturbation negative. Hence $a_2 = 1 - \delta$ for some $\delta > 0$, so $a_2 < 1$.

With dosage rates $a_1 = 1$, $a_2 < 1$, minimizing total time $x_1 + x_2$ subject to $x_1 + a_2 x_2 = 1$ ($x_1, x_2 \geq 0$) yields the unique optimal minimiser $(1, 0)$.

The infinity-norm distance from Jordan's point $(0.3, 0.7)$ to $(1, 0)$ is $\max(|0.3 - 1|, |0.7 - 0|) = 0.7$, which exceeds 0.4. (The later revelation about $(0.6, 0.4)$ being within 0.45 is true, since its distance to $(1, 0)$ is 0.4.)